Distributed transactions and workflows

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Sample chapter for the proposed course plan.

1 Learning outcome

Identify an invariant that spans two services.

Prerequisites: Lectures 06–08.

2 Core idea

A workflow that changes multiple services can stop between steps. An application must decide whether it needs atomic commitment or can expose intermediate states with a recovery or compensation process. Compensation is an application action and is not always an exact undo.

3 Worked scenario

A paid course registration reserves a seat and then requests payment. Payment fails after the seat was reserved.

4 Discussion

What state should the student see, and how should the reservation be released?

5 Recap

Choose coordination mechanisms from the invariant the application must preserve.